Household Appliance Store Database

Business and data model description · Author: Iryna Kapusuz Kutyk

1 Business Description

Business background The Retail Household Appliance Store database is designed to support operations of a store that sells household appliances such as refrigerators, washing machines, ovens, and other electronics. The system will deal with data on products, customers, employees, orders, inventory, and suppliers. It aims to track sales, monitor inventory, manage suppliers etc.

1.1 Problems.

The lack of organized, properly recorded data has negative impact on all business processes, especially in retail sector. We can assume that before a implementing relational database, store faced such minor problems as manual record keeping and partially recorded information, lack of integration between business entities, time consuming processes, security and access control issues.

The Benefits of implementing a database. Project Vision. The database system should let easy and quick access to required information (customers, orders, product description etc.), make a visible link between: customers and orders, orders and employees, products and suppliers and automize tracking of the ordering process. Additionally, the store database will be designed to store, manage, and analyze necessary data with a single, structured system.

2.1 Conceptual Model description

The model has 5 main core business entities such as: customer, order, product, supplier, employee.

Main relationships:

Relationship	Type	Description
Customer → Orders	1:M	One customer can place many orders. Each order is placed by one customer.
Employee → Orders	1:M	One employee can handle many orders. Each order is handled by one employee.
Order → Order_Items	1:M	One order can include multiple order items.
Product → Order_Items	1:M	A product can appear in many order items.
Supplier ↔ Product	M:N	A supplier can supply many products; each product can be supplied by many suppliers. This is implemented via the Product_Supplier table.
Order ↔ Product	M:N	OrderItems (implicit)

2.2 Logical Model description

The database schema is designed to be in 3rd Normal Form (3NF). All tables have primary key, values are atomic. Tables OderItems and ProductSupplier have composite primary key.

All non-key attributes in each table are fully dependent on the whole primary key. For example, in the OrderItems table fields quantity and line_total depend on both order_id and product_id, which together form the composite primary key.

There are no transitive dependancies. In the Product table attributes product_name and price depend only on product_id.

In the Orders table order_date and status depend only order_id.

Using Junction tables:

Product_Supplier table (M:N between Products & Suppliers)

OrderItems table (junction table between Orders & Products).

This avoids redundancy and ensures proper normalization.

Objects:

1.Customers

Column Name	Description	Data Type
customer_id	Unique identifier (PK)	INT (SERIAL)
first_name	Customer's first name	VARCHAR(50)
last_name	Customer's last name	VARCHAR(50)
email	Contact email (UQ)	VARCHAR(100)
phone	Phone number	VARCHAR(30)
address	Customer's address	VARCHAR(150)
created_at	Record creation date	DATE (DEFAULT CURRENT_DATE)

2.Employees

Column Name	Description	Data Type
employee_id	Unique identifier (PK)	INT (SERIAL)
first_name	Employee's first name	VARCHAR(50)
last_name	Employee's last name	VARCHAR(50)
email	Contact email	VARCHAR(100)
position	Job position title	VARCHAR(50)
hired_at	Hire date	DATE (DEFAULT CURRENT_DATE)

3.Products

Column Name	Description	Data Type
product_id	Unique identifier (PK)	INT (SERIAL)
product_name	Name of the product	VARCHAR(100)
brand	Brand name	VARCHAR(50)
category	Product category	VARCHAR(50)
model	Model code	VARCHAR(50)
price	Retail price (must be ≥ 0)	DECIMAL(10,2)
stock	Quantity in stock (≥ 0)	INT
total_value	Calculated total stock value	DECIMAL(12,2)

4.Orders

Column Name	Description	Data Type
order_id	Unique identifier (PK)	INT (SERIAL)
customer_id	Linked to customer (FK)	INT
employee_id	Linked to employee (FK)	INT
order_date	Date of the order	DATE (DEFAULT CURRENT_DATE)
status	Order status ('Pending', etc.)	VARCHAR(50)

5.Order_İtems

Column Name	Description	Data Type
order_id	Linked to order (FK)	INT
product_id	Linked to product (FK)	INT
quantity	Number of units (DEFAULT 1)	INT
unit_price	Price per unit	DECIMAL(10,2)
line_total	Calculated line total	DECIMAL(12,2)
PK	Composite key	(order_id, product_id)

6.Suppliers

Column Name	Description	Data Type
supplier_id	Unique identifier (PK)	INT (SERIAL)
supplier_name	Name of the supplier	VARCHAR(100)
contact_email	Supplier contact email	VARCHAR(100)
phone	Contact phone	VARCHAR(30)

7.Product_Supplier

Column Name	Description	Data Type
product_id	Linked to product (FK)	INT
supplier_id	Linked to supplier (FK)	INT
supply_price	Supply cost	DECIMAL(10,2)
PK	Composite key	(product_id, supplier_id)

Tables and constraints:

Each table was created using CREATE TABLE IF NOT EXISTS to support multiple executions without errors:

Customers: Stores personal details with email marked as UNIQUE.

Employees: Manages staff data with a UNIQUE constraint on email.

Suppliers: Tracks supplier contact info; contact_email is unique.

Products: Catalogs products with product_name + brand + model enforced as UNIQUE, plus price and stock constrained to non-negative values.

Orders: Stores order metadata, linked to Customers and Employees.

OrderItems: Junction table between Orders and Products, with composite PK to prevent duplicates.

Product_Supplier: Junction table for many-to-many relationship between Products and Suppliers, also using composite PK

Date Check: Ensures orders are not before Jan 1, 2024.

Enumerated Order Status: Only allows 'Pending', 'Shipped', or 'Delivered'.

Value Constraints: Prevents negative price or stock.

Uniqueness: Prevents duplicate customers, suppliers, products, or orders using combination constraints like UNIQUE(customer_id, employee_id, order_date).

4**. Data Population (INSERTS)**

Data is added for all tables using INSERT INTO ... ON CONFLICT DO NOTHING, ensuring reusability.

For OrderItems, subqueries are used to resolve foreign key values at runtime based on customer/employee references.

Product_Supplier links each product to a supplier with a price, again avoiding duplication using composite PK logic.

Summary

This recruiting agency database covers all major business processes —customer orders, products description, supplier details, employee data, etc. — in a well-structured, **3NF-compliant** schema as mentioned criteria. The 3NF-compliant schema is achieved as all fields are atomic, every attribute depends on full primary key and there are no transitive dependencies (attributes depend only on PK).

Logical ERD diagram has been attached and represents logically connected relational database system which is ready to solve retail online store challenges. The database is ready to be implemented into work.